

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		- The concentration gradient is maintained / equilibrium is never reached (1) - Oxygen can <u>diffuse</u> into the blood across the <u>whole of the {gill plate / (secondary) lamellae}</u> (1) ignore reference to time <u>More</u> oxygen is absorbed into the blood / blood achieves a <u>higher</u> oxygen saturation / it reaches 80% saturation as compared to 50% (with parallel flow) (1)	3			3		
	(d)			Any two (x1) from {Lungs / Gulping air} enables <u>oxygen</u> to be obtained (when there is insufficient dissolved oxygen in the water) (1) Many capillaries maintain a steep concentration gradient (1) Many (small) air sacs provide a large surface area (1)			2	2		
	(e)			Lungfish and mammals share a common ancestor / shows divergent evolution (1) Reject mammals evolved from lungfish			1	1		
				Question 5 total	9	2	5	16	1	0